Soham Deshpande

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Portfolio: soham-1009.github.io

Professional Summary

AI & ML postgraduate with experience in computer vision, generative AI, and full-stack development. Skilled in Python, PyTorch, REST APIs, Django, FastAPI, JavaScript, and MySQL. Experienced in building AI models and web applications. Strong problem-solving, analytical, communication, and teamwork skills.

Projects

1. InfraNova AI — Thermal Infrared to RGB Satellite Image Translation *(PyTorch, GANs, Computer Vision, FastAPI)*

- Developed a cross-spectral satellite image translation system using a Pix2PixHD-style Global ResNet GAN to synthesize RGB imagery from Landsat 9 TIRS-2 Band 10 + Band 11 inputs.
- Built and evaluated an 11.37M-parameter generator using LSGAN and multi-term loss optimization, achieving 13.74 dB PSNR, 0.4502 SSIM, and 0.3711 YOLOv8 F1@0.25 on 1,259 held-out samples.
- Engineered a FastAPI + React/Vite application with tiled whole-raster inference (128×128 tiles, cosine/Hann blending), processing a 1002×1001 scene in 1.74s at 69.5 tiles/sec, with TTA and CLAHE post-processing.
- Implemented a dual-band Landsat 9 preprocessing and validation pipeline with local percentile normalization, data-quality filtering, paired spatial splits, and robust CHW/HWC input handling for reliable model training and inference.

2. ScamShield AI — Explainable Fraud Detection Platform | *(Python, Scikit-learn, Flask, Chrome Extension, OCR)*

- Engineered a hybrid recruitment-fraud detection engine combining a six-category deterministic rule engine with a regularized TF-IDF + Logistic Regression NLP model, surfacing verbatim matched evidence without LLM API dependencies.
- Built an offline-first Manifest V3 Chrome extension that runs rule evaluation and vector inference locally in JavaScript, with model-weight synchronization and one-click DOM page extraction.
- Developed a containerized Flask REST API for Google Cloud Run with sliding-window rate limiting and Tesseract OCR preprocessing; achieved 100% test coverage across 31 test suites.

3. NoteFlow — Full Stack Notes Management Platform *(Django, MySQL, REST APIs, JavaScript)*

- Developed a full-stack notes management application using Django, MySQL, HTML, CSS, and JavaScript.
- Enforced user authentication, CRUD operations, category-based filtering, and role-based access control.
- Designed relational database schemas using Django ORM and MySQL with support for per-user notes and dashboard analytics.
- Built AJAX-based pin functionality and real-time search capabilities without full page refreshes.

Education

Master of Computer Applications (AI & ML) — Ramdeobaba University, Nagpur | 2025 – Present
Bachelor of Computer Applications — G.H. Raisoni College of Engineering and Management, Nagpur | 2022 – 2025

Skills

Technical Skills:

AI & Machine Learning: GANs (Pix2Pix, Pix2PixHD), YOLOv8, object detection, image-to-image translation
Generative AI: Gemini , ChatGPT, Claude, prompt design, API integration
Programming: Python, Java, SQL
Backend & Web: Django, FastAPI, Flask, REST APIs, HTML, CSS, JavaScript, MySQL, Chrome Extension (Manifest V3)
Data & Analytics: TF-IDF, Logistic Regression, Power BI, Excel, data preprocessing, data validation
Testing & Tools: Git, GitHub, technical documentation, Docker
Core: Data Structures, Algorithms, OOP, DBMS

Soft Skills: Analytical Thinking | Problem Solving | Communication | Research & Documentation | Team Collaboration | Time Management

Certifications

- Completed Japanese N5 course- certificate yet to be received
- Fundamentals of Deep Learning — NVIDIA Certificate of Competency (Aug 2026)
- Google Cloud Career Launchpad — Computing Foundations Track, Google Cloud (Apr 2026)
- Tools of the Trade: Linux and SQL — Google Cybersecurity Certificate, Coursera (Apr 2026)
- Assets, Threats, and Vulnerabilities — Google Cybersecurity Certificate, Coursera (Apr 2026)
- Connect and Protect: Networks and Network Security — Google Cybersecurity Certificate, Coursera (Mar 2026)